ANNIHILATION OF BOUNDED LOCAL COHOMOLOGY OF A COHERENT COMPLEX BY AN IDEAL SUPPORTED AWAY FROM A SPECIALIZATION-STABLE SUBSET

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-local-cohomology/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Dualizing complexes, their dimension functions and the local duality spectral sequence of another chapter. Set up in the cited chapter's section on dimension functions and invoked inside the counted proof at several places (cross-chapter pointers). Supplies the dualizing object, the dimension function and the spectral sequence on which the whole argument runs.
- The earlier annihilator proposition that the counted statement generalises, and the notion of the depth of a complex at a prime. The author's own proposition labelled proposition-annihilator and the cross-chapter definition of depth (pointers). Provide the model proof and the invariant appearing in the second condition.

Source reference. Original reference algebra-lemma-ass-zero: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 2323–2323. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-ass-zero, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original definition definition-depth-complex: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 3138–3138. Navigation pointer only: the counted text refers to the Stacks item definition-depth-complex, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Date: stacks-project:local-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference dualizing-lemma-special-case-local-duality: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 844–844. Navigation pointer only: the counted text refers to the Stacks item dualizing-lemma-special-case-local-duality, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference dualizing-section-dimension-function: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 2276–2276. Navigation pointer only: the counted text refers to the Stacks item dualizing-section-dimension-function, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-local-cohomology-ss: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 1214–1214. Navigation pointer only: the counted text refers to the Stacks item lemma-local-cohomology-ss, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-torsion-change-rings: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 2333–2333. Navigation pointer only: the counted text refers to the Stacks item lemma-torsion-change-rings, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original proposition proposition-annihilator: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 2454–2454. Navigation pointer only: the counted text refers to the Stacks item proposition-annihilator, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference topology-lemma-Noetherian: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 2377–2377. Navigation pointer only: the counted text refers to the Stacks item topology-lemma-Noetherian, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Proposition 0.1. *Let A be a Noetherian ring which has a dualizing complex. Let $T \subset T' \subset \mathrm{Spec}(A)$ be subsets stable under specialization. Let $s \in \mathbf{Z}$. Let K be an object of $D^+_{Coh}(A)$. The following are equivalent*

1. *there exists an ideal $J \subset A$ with $V(J) \subset T'$ such that J annihilates $H^i_T(K)$ for $i \leq s$, and*
2. *for all $\mathfrak{p} \notin T'$, $\mathfrak{q} \in T$ with $\mathfrak{p} \subset \mathfrak{q}$ we have*

$$depth_{A_\mathfrak{p}}(K_\mathfrak{p}) + \dim((A/\mathfrak{p})_\mathfrak{q}) > s$$

Proof. This lemma is the natural generalization of Proposition proposition-annihilator whose proof the reader should read first. Let $\omega_A^\bullet$ be a dualizing complex. Let δ be its

dimension function, see Dualizing Complexes, Section dualizing-section-dimension-function. An important role will be played by the finite A-modules

$$E^i = \mathrm{Ext}^i_A(K, \omega_A^\bullet)$$

For $\mathfrak{p} \subset A$ we will write $H^i_\mathfrak{p}$ to denote the local cohomology of an object of $D(A_\mathfrak{p})$ with respect to $\mathfrak{p}A_\mathfrak{p}$. Then we see that the $\mathfrak{p}A_\mathfrak{p}$-adic completion of

$$(E^i)_\mathfrak{p} = \mathrm{Ext}^{\delta(\mathfrak{p})+i}_{A_\mathfrak{p}}(K_\mathfrak{p}, (\omega_A^\bullet)_\mathfrak{p}[-\delta(\mathfrak{p})])$$

is Matlis dual to

$$H^{-\delta(\mathfrak{p})-i}_\mathfrak{p}(K_\mathfrak{p})$$

by Dualizing Complexes, Lemma dualizing-lemma-special-case-local-duality. In particular we deduce from this the following fact: an ideal $J \subset A$ annihilates $(E^i)_\mathfrak{p}$ if and only if J annihilates $H^{-\delta(\mathfrak{p})-i}_\mathfrak{p}(K_\mathfrak{p})$.

Set $T_n = \{\mathfrak{p} \in T \mid \delta(\mathfrak{p}) \leq n\}$. As δ is a bounded function, we see that $T_a = \emptyset$ for $a \ll 0$ and $T_b = T$ for $b \gg 0$.

Assume (2). Let us prove the existence of J as in (1). We will use a double induction to do this. For $i \leq s$ consider the induction hypothesis IH_i: $H^a_T(K)$ is annihilated by some $J \subset A$ with $V(J) \subset T'$ for $a \leq i$. The case IH_i is trivial for i small enough because K is bounded below.

Induction step. Assume IH_{i-1} holds for some $i \leq s$. Pick J' with $V(J') \subset T'$ annihilating $H^a_T(K)$ for $a \leq i-1$ (the induction hypothesis guarantees we can do this). We will show by descending induction on n that there exists an ideal J with $V(J) \subset T'$ such that the associated primes of $JH^i_T(K)$ are in T_n. For $n \ll 0$ this implies $JH^i_T(K) = 0$ (Algebra, Lemma algebra-lemma-ass-zero) and hence IH_i will hold. The base case $n \gg 0$ is trivial because $T = T_n$ in this case and all associated primes of $H^i_T(K)$ are in T.

Thus we assume given J with the property for n. Let $\mathfrak{q} \in T_n$. Let $T_\mathfrak{q} \subset \mathrm{Spec}(A_\mathfrak{q})$ be the inverse image of T. We have $H^j_T(K)_\mathfrak{q} = H^j_{T_\mathfrak{q}}(K_\mathfrak{q})$ by Lemma lemma-torsion-change-rings. Consider the spectral sequence

$$H^p_\mathfrak{q}(H^q_{T_\mathfrak{q}}(K_\mathfrak{q})) \Rightarrow H^{p+q}_\mathfrak{q}(K_\mathfrak{q})$$

of Lemma lemma-local-cohomology-ss. Below we will find an ideal $J'' \subset A$ with $V(J'') \subset T'$ such that $H^i_\mathfrak{q}(K_\mathfrak{q})$ is annihilated by J'' for all $\mathfrak{q} \in T_n \setminus T_{n-1}$. Claim: $J(J')^i J''$ will work for $n-1$. Namely, let $\mathfrak{q} \in T_n \setminus T_{n-1}$. The spectral sequence above defines a filtration

$$E^{0,i}_\infty = E^{0,i}_{i+2} \subset \ldots \subset E^{0,i}_3 \subset E^{0,i}_2 = H^0_\mathfrak{q}(H^i_{T_\mathfrak{q}}(K_\mathfrak{q}))$$

The module $E^{0,i}_\infty$ is annihilated by J''. The subquotients $E^{0,i}_j/E^{0,i}_{j+1}$ for $i+1 \geq j \geq 2$ are annihilated by J' because the target of $d^{0,i}_j$ is a subquotient of

$$H^j_\mathfrak{q}(H^{i-j+1}_{T_\mathfrak{q}}(K_\mathfrak{q})) = H^j_\mathfrak{q}(H^{i-j+1}_T(K)_\mathfrak{q})$$

and $H^{i-j+1}_T(K)_\mathfrak{q}$ is annihilated by J' by choice of J'. Finally, by our choice of J we have $JH^i_T(K)_\mathfrak{q} \subset H^0_\mathfrak{q}(H^i_T(K)_\mathfrak{q})$ since the non-closed points of $\mathrm{Spec}(A_\mathfrak{q})$ have higher δ values. Thus $\mathfrak{q}$ cannot be an associated prime of $J(J')^i J'' H^i_T(K)$ as desired.

By our initial remarks we see that J'' should annihilate

$$(E^{-\delta(\mathfrak{q})-i})_\mathfrak{q} = (E^{-n-i})_\mathfrak{q}$$

for all $\mathfrak{q} \in T_n \setminus T_{n-1}$. But if J'' works for one $\mathfrak{q}$, then it works for all $\mathfrak{q}$ in an open neighbourhood of $\mathfrak{q}$ as the modules E^{-n-i} are finite. Since every subset

of $\mathrm{Spec}(A)$ is Noetherian with the induced topology (Topology, Lemma topology-lemma-Noetherian), we conclude that it suffices to prove the existence of J'' for one $\mathfrak{q}$.

Since the ext modules are finite the existence of J'' is equivalent to

$$\mathrm{Supp}(E^{-n-i}) \cap \mathrm{Spec}(A_\mathfrak{q}) \subset T'.$$

This is equivalent to showing the localization of E^{-n-i} at every $\mathfrak{p} \subset \mathfrak{q}$, $\mathfrak{p} \notin T'$ is zero. Using local duality over $A_\mathfrak{p}$ we find that we need to prove that

$$H^{i+n-\delta(\mathfrak{p})}_\mathfrak{p}(K_\mathfrak{p}) = H^{i-\dim((A/\mathfrak{p})_\mathfrak{q})}_\mathfrak{p}(K_\mathfrak{p})$$

is zero (this uses that δ is a dimension function). This vanishes by the assumption in the lemma and $i \leq s$ and our definition of depth in Definition definition-depth-complex.

To prove the converse implication we assume (2) does not hold and we work backwards through the arguments above. First, we pick a $\mathfrak{q} \in T$, $\mathfrak{p} \subset \mathfrak{q}$ with $\mathfrak{p} \notin T'$ such that

$$i = \mathrm{depth}_{A_\mathfrak{p}}(K_\mathfrak{p}) + \dim((A/\mathfrak{p})_\mathfrak{q}) \leq s$$

is minimal. Then $H^{i-\dim((A/\mathfrak{p})_\mathfrak{q})}_\mathfrak{p}(K_\mathfrak{p})$ is nonzero by the our definition of depth in Definition definition-depth-complex. Set $n = \delta(\mathfrak{q})$. Then there does not exist an ideal $J \subset A$ with $V(J) \subset T'$ such that $J(E^{-n-i})_\mathfrak{q} = 0$. Thus $H^i_\mathfrak{q}(K_\mathfrak{q})$ is not annihilated by an ideal $J \subset A$ with $V(J) \subset T'$. By minimality of i it follows from the spectral sequence displayed above that the module $H^i_T(K)_\mathfrak{q}$ is not annihilated by an ideal $J \subset A$ with $V(J) \subset T'$. Thus $H^i_T(K)$ is not annihilated by an ideal $J \subset A$ with $V(J) \subset T'$. This finishes the proof of the proposition. $\square$